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Q#01:-

a) $P(X=0) = \frac{\binom{2}{0}\binom{18}{5}}{\binom{20}{5}} = \frac{51}{92} \approx 0.5543$

b) $P(X \geq 1) = 1 - P(X=0) = 1 - \frac{51}{92} = \frac{41}{92} \approx 0.4457$

c) only 2 defected; $X = 0, 1, 2$ ∴ at most = 2

$P(X \leq 2) = 1$

d) $\mu = E(X) = n \cdot \frac{K}{N} = 5 \cdot \frac{2}{20} = \frac{1}{2} = 0.5$

$V(X) = \frac{nK}{N}\left(1 - \frac{K}{N}\right)\frac{N-n}{N-1} = \frac{5 \cdot 2}{20} \cdot \frac{18}{20} \cdot \frac{15}{19}$

$= \frac{27}{76} \approx 0.3553$

Q#02:-

a) $P(X=10) = (1-0.1)^9 (0.1) \approx 0.03874$

b) $P(X=5) = (1-0.03)^4 (0.03) \approx 0.02656$

CamScanner

Q#03:-

a) $E(X) = 1/p = 1/0.15 = 20/3 \approx 6.67$

expected about 7 people

b) $P(4 \le X \le 8) = \sum_{k=4}^{8} (0.85)^{k-1}(0.15)$

$= 0.15\left[0.85^3 + 0.85^4 + 0.85^5 + 0.85^6 + 0.85^7\right]$

$= 0.341634475 \approx 0.3416$

Q#04:-

a) $P = \binom{5}{1}(0.25)^1(0.75)^4(0.25) \approx 0.0989$

$p = 0.25$

$E(X) = r/p = 2/0.25 = 8$

$V(X) = \frac{r(1-p)}{p^2} = \frac{2(1-0.25)}{0.25^2} = 24$

b) $p = 0.1$; $E(X) = r/p = 3/0.1 = 30$

average 30 calls

$P(X = 25) = \binom{24}{2}(0.1)^2(0.9)^{22}(0.1)$

≈ 0.0272

Q#05:-

a) $P(X=0) = e^{-0.5}$ | ∵ for 5 independent disks $(e^{-0.5})^5 = e^{-2.5}$

$P = e^{-2.5} \approx 0.0821$

b) $T \approx \text{Poisson}(5 \times 0.5) = \text{Poisson}(2.5)$

$P(T=2) = \frac{e^{-2.5}(2.5)^2}{2!} \approx 3.12\, e^{-2.5}$

≈ 0.2565

c) $P(T \geq 2) = 1 - P(T=0) - P(T=1) \because 1 - P(T<2)$

$P(T=0) = e^{-2.5}$, $P(T=1) = e^{-2.5}(2.5)$

$P(T \geq 2) = 1 - e^{-2.5} - 2.5e^{-2.5} \approx 0.7127$

Q#06:-

a) $\sum_{x=1}^{\infty} P(X=x) = 1 \quad \because P(X=x) = pq^{x-1}$

$\sum_{x=1}^{\infty} pq^{x-1} = p\sum_{x=1}^{\infty} q^{x-1}$ let $r = x-1$

where $r = 0, 1, 2, \ldots$

$= p\sum_{r=0}^{\infty} q^r$

Geometric Series

$$\because \sum_{r=0}^{\infty} q^r = \frac{1}{1-q}$$

$$p\sum_{r=0}^{\infty} q^r = p \cdot \frac{1}{1-q} \quad \because 1-q = p$$

$$p \cdot \frac{1}{p} = 1$$

b) $$E(X) = \sum_{x=1}^{\infty} x \cdot P(X=x) = \sum_{x=1}^{\infty} x p q^{x-1} = p\sum_{x=1}^{\infty} x \cdot q^{x-1}$$

$$\because \sum_{x=1}^{\infty} x \cdot q^{x-1} = \frac{1}{(1-q)^2} \; ; \; E(X) = p \cdot \frac{1}{(1-q)^2}$$

$$\because 1-q = p \; ; \; E(X) = p \cdot \frac{1}{p^2} = 1/p$$

c) ~~$q/p^2 \Rightarrow$~~ $V(X) = q/p^2$

$$V(X) = E(X^2) - (E(X))^2 \Rightarrow E(X) = 1/p$$

$$E(X^2) = \sum_{x=1}^{\infty} x^2 \cdot P(X=x) = \sum_{x=1}^{\infty} x^2 \cdot p q^{x-1} = p\sum_{x=1}^{\infty} x^2 \cdot q^{x-1}$$

~~$E(X^2) = p \cdot \frac{1+q}{(1-q)^3}$~~ $\because \sum_{x=1}^{\infty} x^2 \cdot q^{x-1} = \frac{1+q}{(1-q)^3}$

$$E(X^2) = p \cdot \frac{(1+q)}{(1-q)^3} = p \cdot \frac{(1+q)}{p^3} = \frac{1+q}{p^2}$$

$$V(X) = \frac{1+q}{p^2} - \left(\frac{1}{p}\right)^2 = \frac{q}{p^2}$$

d) $M_x(t) = E(e^{tx})$ $\because$ PMF $= M_x(t) = \sum_{x=1}^{\infty} e^{tx} pq^{x-1}$

$\because e^{tx} = e^t (e^t)^{x-1}$; $M_x(t) = pe^t \sum_{x=1}^{\infty} (qe^t)^{x-1}$

$$\sum_{x=1}^{\infty} (qe^t)^{x-1} = \sum_{r=0}^{\infty} (qe^t)^r = \frac{1}{1-qe^t}$$

Mean using MGF $\because F(X) = M'_x(0)$

$M_x(t) = \frac{pe^t}{1-qe^t}$, Differentiate $M'_x(t)$

$$M'_x(t) = \frac{pe^t(1-qe^t) - pe^t(-qe^t)}{(1-qe^t)^2} = \frac{pe^t(1-qe^t+qe^t)}{(1-qe^t)^2}$$

$= \frac{pe^t}{(1-qe^t)^2}$; put $t=0$ $M'x(0) = \frac{p}{(1-q)^2}$

$\frac{p}{p^2} = \frac{1}{p}$ $E(X) = \frac{1}{p}$

Variance Using MGF

$$V(X) = E(X^2) - E(X)^2$$

$$E(X^2) = M''_X(0)$$

$$M'_X(t) = pe^t (1-qe^t)^{-2}$$

$$M''_X(t) = pe^t (1-qe^t)^{-2} + pe^t \cdot 2qe^t (1-qe^t)^{-3}$$

$$= \frac{pe^t}{(1-qe^t)^2} + \frac{2pqe^{2t}}{(1-qe^t)^3} \quad ; \text{put } t=0$$

$$M''_X(0) = \frac{p}{(1-q)^2} + \frac{2pq}{(1-q)^3} \quad \because 1-q=p$$

$$\frac{p}{p^2} + \frac{2pq}{p^3} = \frac{q}{p^2}$$

$\because p+q=1$
$\hookrightarrow p+2q=1+q$

e) $K_X(t) = \ln M_X(t)$

$$M_X(t) = \frac{pe^t}{1-qe^t} \Rightarrow K_X(t) = \ln\left(\frac{pe^t}{1-qe^t}\right)$$

$$= \ln(pe^t) - \ln(1-qe^t)$$

Mean using CGF

$$E(X) = K'_X(0)$$

$$\frac{d}{dt}\left[\ln P + t + \ln(1-qe^t)\right]$$

$$= 0 + 1 - \left(\frac{-qe^t}{1-qe^t}\right) = 1 + \frac{qe^t}{1-qe^t}$$

At $t=0$; $K'_X(0) = 1 + \frac{q}{1-q} = 1 + \frac{q}{p} = \frac{1}{p}$

Variance Using CGF:

$$V(X) = E(X^2) - (E(X))^2 = K''_X(0) - (K'_X(0))^2$$

$$K'_X(t) = 1 + \frac{qe^t}{1-qe^t}$$

$$K''_X(t) = \frac{qe^t(1-qe^t) - qe^t(-qe^t)}{(1-qe^t)^2} = \frac{qe^t}{(1-qe^t)^2}$$

Put $t=0$ $\quad K''_X(0) = \frac{q}{(1-q)^2} = \frac{q}{p^2}$

~~Q#08:-~~

Q#07:-

a) $$\int_0^1 f(x)\,dx = 1 \Rightarrow \int_0^1 \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)+\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$$

$$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx = \int_0^1 x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$$

$$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} \cdot \frac{\Gamma(\alpha)\cdot\Gamma(\beta)}{\Gamma(\alpha+\beta)} = 1$$

b) $$E(X) = \int_0^1 x \cdot f(x)\,dx$$

$$E(X) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} \int_0^1 x^{\alpha}\cdot(1-x)^{\beta-1}\,dx$$

$$\therefore \int_0^1 x^{\alpha}(1-x)^{\beta-1}\,dx = B(\alpha+1,\beta) = \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}$$

$\because \Gamma(\alpha+1) = \alpha\Gamma(\alpha)$

$$E(X) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+1)\cdot\Gamma(\beta)}{\Gamma(\alpha+\beta+1)} \quad \because \Gamma(\alpha+1) = \alpha\Gamma(\alpha)$$

$$E(X) = \frac{\Gamma(\alpha+\beta)\cdot\alpha\Gamma(\alpha)\cdot\Gamma(\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)(\alpha+\beta)\Gamma(\alpha+\beta)} = \frac{\alpha}{\alpha+\beta}$$

c) $V(X) = E(X^2) - (E(X))^2 \quad \because E(X) = \frac{\alpha}{\alpha+\beta}$

$$E(X^2) = \int_0^1 x^2 \cdot f(x)\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \int_0^1 x^{\alpha+1}(1-x)^{\beta-1}\,dx$$

$$\because \int_0^1 x^{\alpha+1}(1-x)^{\beta-1}\,dx = B(\alpha+2,\beta) = \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)}$$

$$E(X^2) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} \cdot \frac{\Gamma(\alpha+2)\Gamma(\beta)}{\Gamma(\alpha+\beta+2)} \quad \because \Gamma(\alpha+2) = (\alpha+1)\alpha\Gamma(\alpha)$$
$$\because \Gamma(\alpha+\beta+2) = (\alpha+\beta+1)(\alpha+\beta)\cdot\Gamma(\alpha+\beta)$$

$$= \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)}$$

$$V(X) = \frac{\alpha(\alpha+1)}{(\alpha+\beta)(\alpha+\beta+1)} - \left(\frac{\alpha}{\alpha+\beta}\right)^2 = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$$

d) $M_X(t) = E(e^{tx}) = \int_0^1 e^{tx} \cdot f(x)\,dx$

$$M_X(t) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} \int_0^1 e^{tx} \cdot x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$e^{tx} = \sum_{n=0}^{\infty} \frac{(tx)^n}{n!}$$

$$M_X(t) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} \int_0^1 \left(\sum_{n=0}^{\infty} \frac{t^n x^n}{n!}\right) x^{\alpha-1}(1-x)^{\beta-1}\,dx$$

$$= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} \sum_{n=0}^{\infty} \frac{t^n}{n!} \int_0^1 x^{\alpha+n-1}(1-x)^{\beta-1}\,dx$$

$$B(\alpha+n,\beta) = \frac{\Gamma(\alpha+n)\Gamma(\beta)}{\Gamma(\alpha+\beta+n)}$$

$$M_X(t) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)} \sum_{n=0}^{\infty} \frac{t^n}{n!} \cdot \frac{\Gamma(\alpha+n)\Gamma(\beta)}{\Gamma(\alpha+\beta+n)}$$

$$M_X(t) = \sum_{n=0}^{\infty} \frac{\Gamma(\alpha+\beta)\Gamma(\alpha+n)t^n}{\Gamma(\alpha)\,\Gamma(\alpha+\beta+n)\,n!}$$

Q#08:-

a) $$\int_{-\infty}^{\infty} f(x)\,dx = 1 \Rightarrow \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx \; ; \; let\; z = \frac{x-\mu}{\sigma}$$

$$\boxed{dx = \sigma\, dz}, \; then \; \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2} \sigma\, dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-z^2/2}\, dz \Rightarrow \int_{-\infty}^{\infty} e^{-z^2/2}\, dz = \sqrt{2\pi}$$

$$\frac{1}{\sqrt{2\pi}} \cdot \sqrt{2\pi} = 1$$

b) $$E(X) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

$$= \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$E(X) = \int_{-\infty}^{\infty} (\mu + \sigma z) \frac{1}{\sqrt{2\pi}} \cdot e^{-z^2/2}\, dz$$

$$E(X) = \mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz + \sigma \int_{-\infty}^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 1 \quad \text{and} \quad \int_{-\infty}^{\infty} z \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 0$$

$$E(X) = \mu(1) + \sigma(0) = \mu$$

c) $V(X) = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx$

$$= \int_{-\infty}^{\infty} (x-\mu)^2 \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

Let $z = \frac{x-\mu}{\sigma}$, $x - \mu = \sigma z$, $dx = \sigma\, dz$

$$= \int_{-\infty}^{\infty} (\sigma z)^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = \sigma^2 \int_{-\infty}^{\infty} z^2 \cdot \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz$$

$\because \int_{-\infty}^{\infty} z^2 \frac{1}{\sqrt{2\pi}} e^{-z^2/2} dz = 1$

$V(X) = 1$

d) $M_x(t) = e^{\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)}$

$$M_x(t) = E(e^{tx}) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{\left[tx - \frac{(x-\mu)^2}{2\sigma^2}\right]} dx$$

$$\because tx - \frac{(x-\mu)^2}{2\sigma^2} = -\frac{1}{2\sigma^2}(x - \mu - \sigma^2 t)^2 + \mu t + \frac{1}{2}\sigma^2 t^2$$

$$M_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \cdot \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2\sigma^2}(x-\mu-\sigma^2 t)^2} dx$$

$\because$ But that integral is 1.

$$M_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

Mean using MGF

$E(X) = M'_x(0)$ $\because M_{X(t)} = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

$M'_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}(\mu + \sigma^2 t)$; Now put $t=0$

$M'_x(0) = e^0(\mu + 0) = \mu$

Variance using MGF

$E(X^2) = M''_x(0)$

$M'_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}(\mu + \sigma^2 t)$

$M''_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}(\mu + \sigma^2 t)^2 + e^{\mu t + \frac{1}{2}\sigma^2 t^2}\sigma^2$

$M'_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}\left[(\mu + \sigma^2 t)^2 + \sigma^2\right]$

Put $t = 0$

$M''_x(0) = \mu^2 + \sigma^2$

$V(X) = E(X^2) - (E(X))^2 = (\mu^2 + \sigma^2) - \mu^2 = \sigma^2$

e) $K_x(t) = \ln M_x(t)$

$M_x(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$

$K_x(t) = \mu t + \frac{1}{2}\sigma^2 t^2$

Mean using CGF

$E(X) = K'_x(0)$

$K'_x(t) = \mu + \sigma^2 t$; Put $t = 0$

$K'_x(0) = \mu$

$E(X) = \mu$

Variance Using CGF

$V(X) = K''_x(0)$

$K''_x(t) = \sigma^2$

$K''_x(0) = \sigma^2$

$V(X) = \sigma^2$

Q#09:-

a) $W \leq 10$; $P(W \leq 10) = \frac{10}{30} = \frac{1}{3}$

b) $W \geq 15$; $P(W \geq 15) = \frac{30-15}{30} = \frac{1}{2}$

Q#10:-

a) $P(X < 153)$

$$Z = \frac{153 - 150.3}{5} = 0.54$$

$P(X < 153) = P(Z < 0.54) \approx 0.7054$

b) $P(X > 158)$

$$Z = \frac{158 - 150.3}{5} = 1.54$$

$P(X > 158) = P(Z > 1.54) = 1 - P(Z < 1.54)$

$P(Z < 1.54) \approx 0.9382$

$P(X > 158) = 1 - 0.9382 = 0.0618$

c) $P(150 < X < 158)$

for $X = 150$

$$Z = \frac{150 - 150.3}{5} = -0.06$$

for $X = 158$

$$Z = \frac{158 - 150.3}{5} = 1.54$$

$P(150 < X < 158) = P(-0.06 < Z < 1.54)$

$= P(Z < 1.54) - P(Z < -0.06)$

$P(Z < 1.54) \approx 0.9382$

$P(Z < -0.06) \approx 0.4761$

$P(150 < X < 158) = 0.9382 - 0.4761$

$= 0.4621$